

Bhalaji Yadav Kantepalle

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PROFESSIONAL SUMMARY

Chemical Engineer (M.S.) with 3+ years of experience bridging **Materials Science** and **Technical Operations**. Specializes in building digital frameworks that make manufacturing **compliant**, **reliable**, and **scalable**. Proven track record of automating R&D metrology (Python/OpenCV) and directing FDA-regulated Quality Systems (SAP/cGMP). Expert in translating complex material data into actionable engineering decisions.

EDUCATION

Virginia Commonwealth University (VCU)

M.S., Chemical & Life Science Engineering | **GPA: 3.62**

Focus: Fracture Mechanics, Optical Metrology, Computational Fluid Dynamics (CFD), Scientific Computing.

Richmond, VA

Exp. Dec 2026

Award: Graduate Assistantship (Fall 2025) – Full Tuition Waiver & Stipend.

Amrita University

B.Tech., Chemical Engineering

Coimbatore, India

Jun 2022

TECHNICAL SKILLS

Engineering Operations: FDA Compliance, SAP ERP Validation, Root Cause Analysis (RCA), CAPA
Materials Science: Fracture Mechanics, Adhesion Testing (ASTM D2724), Rheology, Optical Microscopy, Polymer Physics
Data Automation: Python (Pandas, OpenCV, Scikit-learn), Docker, Tableau, SQL, Image Processing
Lab Instruments: Tensile Testers (TA.XTplusC Texture Analyser), Rheometers, NMR Spectroscopy, UV-Vis

EXPERIENCE

VCU Soft Functional Materials Lab

Graduate Researcher & Metrology Lead

Richmond, VA

Sep 2024 – Present

- **Integrated Optical Metrology System:** Engineered a high-throughput Python/OpenCV pipeline to quantify mechanochromic response, synchronizing sub-pixel texture tracking with rigorous **CIE L*a*b*** colorimetry.
- **Automated Failure Analysis:** Developed a fracture signal analysis protocol that eliminates "false positive" adhesion metrics in soft wearables, reducing data processing time by **>80%**.
- **Grant Success:** Drafted a successful **\$30,000 research grant**, synthesizing technical data to substantiate the clinical viability of smart textile sensors.

Kreative Organics Pvt. Ltd.

Technical Project Manager

Hyderabad, India

May 2023 – May 2024

- **FDA Compliance Leadership:** Managed the **SAP ERP Re-qualification** lifecycle, partnering with external consultants to execute OQ/PQ protocols that ensured full **cGMP/FDA audit readiness**.
- **Supply Chain Automation:** Spearheaded a digital transformation initiative using Python and Docker to standardize chemical nomenclature, reducing market research timelines by **70%**.
- **Operational Excellence:** Directed CAPA and Root Cause Analysis (RCA) initiatives for manufacturing deviations, implementing SOP updates that reduced process variability.

Technical Project Intern

Oct 2022 – Apr 2023

- **Data Strategy:** Spearheaded and containerized a standardized data protocol (Docker) to eliminate naming inconsistencies in global trade logs, driving a 90% uplift in retrieval accuracy.

PUBLICATIONS & ACHIEVEMENTS

Journal Article: I. Caloian, J. Trapp, **B. Y. Kantepalle**, et al. "Mechanical Properties of **Dual-Layer Electrospun Fiber Mats**". *Polymers* 17(13), 1777 (2025). [doi:10.3390/polym17131777]

Selected Participant: 1st National Neutron Scattering School, Oak Ridge National Laboratory (Sep 2025).

Poster Presentation: "Adhesives for Personalized Wearable Devices" (ACS Fall 2025, Washington D.C.).